Lecture 5

EWMA
(Smoothing Out the Noise for Better Predictions)

Exponentially Weighted Moving Average (EWMA) is a technique used to **smooth data** by giving exponentially decreasing weights to past observations. It is commonly used in time-series analysis, optimization, and adaptive learning rates.

- **Smoothing data** means reducing noise or fluctuations in a dataset while preserving the overall trend. It helps in identifying patterns by eliminating short-term variations that might obscure important insights.

Exponentially Weighted Moving Average (EWMA):

$$\boldsymbol{EWMA_t = \beta x_t + (1 - \beta)\, EWMA_{t-1}}$$

- $EWMA_t$ is the EWMA at time t.
- x_t is the data point at time t (current data point).
- $EWMA_{t-1}$ is the EWMA at the previous time step $(t - 1)$
- $\boldsymbol{\beta}$ is the **smoothing parameter (between 0 and 1)** that controls the **weight given to the most recent data point**.

EWMA

- For $t = 1$, let $EWMA_1 = \beta x_1$ (any initial value), **assuming $\boldsymbol{EWMA_0 = 0}$**
- For $t = 2$, let $EWMA_2 = \beta x_2 + (1-\beta)\, EWMA_1 = \beta x_2 + (1-\beta)\, \beta x_1$
- For $t = 3$, let $EWMA_3 = \beta x_3 + (1-\beta)\, EWMA_2$

  $= \beta x_3 + (1-\beta)\, [\beta x_2 + (1-\beta)\, EWMA_1] = \beta x_3 + (1-\beta)\, [\beta x_2 + (1-\beta)\beta x_1]$

  $= \beta x_3 + \beta(1-\beta) x_2 + \beta(1-\beta)^2 x_1$
- For $t = 4$, let $EWMA_4 = \beta x_4 + (1-\beta)\, EWMA_3$

  $= \beta x_4 + (1-\beta)[\beta x_3 + \beta(1-\beta) x_2 + \beta(1-\beta)^2 x_1]$

  $= \beta x_4 + \beta(1-\beta) x_3 + \beta(1-\beta)^2 x_2 + \beta(1-\beta)^3 x_1$
- In general:
- $EWMA_t$ is a weighted sum of the current data point x_t and past data points $x_{t-1,} x_{t-2,\dots}$ with weights decreasing exponentially..

$$EWMA_t = \beta x_t + \beta(1-\beta) x_{t-1} + \beta(1-\beta)^2 x_{t-2} + \cdots \beta(1-\beta)^k x_{t-k} + \cdots + \beta(1-\beta)^{t-2} x_2 + \beta(1-\beta)^{t-1} x_1$$

  - *The index k represents how far back a particular term is from the current time step t.*
- Equivalently:

$$EWMA_t = \sum_{k=0}^{t-1} \beta(1-\beta)^k x_{t-k}$$

EWMA

$$EWMA_t = \sum_{k=0}^{t-1} \beta(1-\beta)^k x_{t-k}$$

Exponentially decay weights:

- For data point x_{t-k}, weight $W_k = \beta(1-\beta)^k$
- Weight of $x_t = \beta$ (i.e., $k = 0$)
- Weight of $x_{t-1} = \beta(1-\beta)$ (i.e., $k = 1$)
- Weight of $x_{t-2} = \beta(1-\beta)^2$ (i.e., $k = 2$)
- Weight for $x_1 = \beta(1-\beta)^{t-1}$ (i.e., $k = t-1$)
- Total weights: $\sum_{k=0}^{t-1} \beta\,(1-\beta)^k$ =?
  - This is a geometric series with **first term β** and **common ratio $(1-\beta)$**
  - Using the sum formula for a geometric series:
    - $S = a\frac{1-r^n}{1-r}$
    - $\sum_{k=0}^{t-1} \beta\,(1-\beta)^k = \beta\frac{1-(1-\beta)^t}{1-(1-\beta)}$
    - $\sum_{k=0}^{t-1} \beta\,(1-\beta)^k = \beta\frac{1-(1-\beta)^t}{\beta} = 1-(1-\beta)^t \approx 1$ (as $t \to \infty$)
- **It means all past data is accounted for, but exponentially less for older observations. Thus EWMA forms a proper weighted average.**

EWMA

EWMA for $x^{(k)}$ at k^{th} iteration is:
$EWMA^{(k)} = \beta x^{(k)} + (1-\beta)\, EWMA^{(k-1)}$

- Let $\beta = 0.9$
- $EWMA^{(k)} = 0.9x^{(k)} + (1-0.9)\, EWMA^{(k-1)} = 0.9x^{(k)} + 0.1EWMA^{(k-1)}$
  $= 0.9x^{(k)} + 0.1\left[0.9x^{(k-1)} + (1-0.9)\, EWMA^{(k-2)}\right]$
  $= 0.9x^{(k)} + 0.09x^{(k-1)} + 0.01EWMA^{(k-2)}$
  $= 0.9x^{(k)} + 0.09x^{(k-1)} + 0.01\left[0.9x^{(k-2)} + (1-0.9)\, EWMA^{(k-3)}\right]$
  $= 0.9x^{(k)} + 0.09x^{(k-1)} + 0.009x^{(k-2)} + 0.001EWMA^{(k-3)}$
- In general, after n iteration:
- $EWMA^{(k)} = 0.9x^{(k)} + 0.09x^{(k-1)} + 0.009x^{(k-2)} + \cdots + \cdots (0.1)^n x^{(k-n+1)} + (0.1)^n EWMA^{(k-n)}$

For any $\beta \in [0,1]$, the weights in EWMA decreases, but the rate at which they decrease will vary.
$\boldsymbol{\beta \to 1}$: The weights decay rapidly, so recent data has a much larger impact compared to older data.
$\boldsymbol{\beta \to 0}$: The weights decay slowly, so older data retains more influence, leading to a smoother EWMA that reacts more slowly to recent changes.

- Let $\beta = 0.3$
- $EWMA^{(k)} = 0.3x^{(k)} + (1-0.3)\, EWMA^{(k-1)} = 0.3x^{(k)} + 0.7EWMA^{(k-1)}$
  $= 0.3x^{(k)} + 0.7\left[0.3x^{(k-1)} + (1-0.3)\, EWMA^{(k-2)}\right]$
  $= 0.3x^{(k)} + 0.21x^{(k-1)} + 0.49EWMA^{(k-2)}$
  $= 0.3x^{(k)} + 0.21x^{(k-1)} + 0.49\left[0.3x^{(k-2)} + (1-0.3)\, EWMA^{(k-3)}\right]$
  $= 0.3x^{(k)} + 0.21x^{(k-1)} + 0.147x^{(k-2)} + 0.343EWMA^{(k-3)}$
- In general, after n iteration:
- $EWMA^{(k)} = 0.3\, x^{(k)} + 0.21x^{(k-1)} + 0.147x^{(k-2)} + 0.7^n x^{k-n+1} + \cdots + (0.7)^n EWMA^{(k-n)}$

RMSProp

Adagrad

%Inputs

$L(W)$: $\mathbb{R}^n \to \mathbb{R}$, a smooth loss function

$W^{(0)} \in \mathbb{R}^n$ The initial weight parameters

$G^{(0)} = 0_n \in \mathbb{R}^n$: Archive for Accumulated square gradient

$\alpha \in \mathbb{R}$: Learning rate

ε: a small positive value

%Output: $W^* \in \mathbb{R}^n$ optimal weight vector

Begin

$k \leftarrow 1$

While (stopping criteria $\left\|\nabla_W L\left(W^{(k)}\right)\right\| > \varepsilon$) **do**

- Compute gradient $\nabla_W L\left(W^{(k)}\right)$
- $G^{(k)} = G^{(k-1)} + \left(\nabla_W L\left(W^{(k)}\right)\right)^2$ % Accumulate square gradient
- $W^{(k+1)} = W^{(k)} - \frac{\alpha}{\sqrt{G^{(k)}+\varepsilon}} \nabla_W L\left(W^{(k)}\right)$ %Update position
- $k = k + 1$;

EndWhile

End

RMSProp

%Inputs

$L(W)$: $\mathbb{R}^n \to \mathbb{R}$, a smooth loss function

$W^{(0)} \in \mathbb{R}^n$ The initial weight parameters

$G^{(0)} = 0_n \in \mathbb{R}^n$: Archive for Accumulated square gradient

$\alpha \in \mathbb{R}$: Learning rate

ε: a small positive value

$\beta \in [0,1]$: Smoothing parameter for EWMA (most preferable for DL models, value $\beta = 0.9$)

%Output: $W^* \in \mathbb{R}^n$ optimal weight vector

Begin

$k \leftarrow 1$

While (stopping criteria $\left\|\nabla_W L\left(W^{(k)}\right)\right\| > \varepsilon$) **do**

- Compute gradient $\nabla_W L\left(W^{(k)}\right)$
- $G^{(k)} = \beta G^{(k-1)} + (1-\beta)\left(\nabla_W L\left(W^{(k)}\right)\right)^2$ % Accumulate square gradient by **EWMA**
- $W^{(k+1)} = W^{(k)} - \frac{\alpha}{\sqrt{G^{(k)}+\varepsilon}} \nabla_W L\left(W^{(k)}\right)$ %Update position
- $k = k + 1$;

EndWhile

End

- RMSProp (Root mean square propagation):
- Root mean square (RMS): the square root of the average (EWMA) of the squared gradients

RMSProp for 2 D feature space

RMSProp for 2 D feature space

%Inputs

$L(W)$: $\mathbb{R}^2 \to \mathbb{R}$, a smooth loss function

$W^{(0)} = (w_1^{(0)}, w_2^{(0)}) \in \mathbb{R}^2$ The initial weight parameters

$G^{(0)} = \left(g_1^{(0)}, g_2^{(0)}\right) = (0,0) \in \mathbb{R}^2$: Accumulated square gradient

$\alpha \in \mathbb{R}$: Learning rate

ε: a small positive value

$\beta \in [0,1]$: Smoothing parameter for EWMA (most preferable for DL models, value $\beta = 0.9$)

%Output: $W^* \in \mathbb{R}^2$ optimal weight vector

Begin

$k \leftarrow 1$

While (stopping criteria $\left\|\nabla L\left(w_1^{(k)}, w_2^{(k)}\right)\right\| > \varepsilon$) **do**

- Compute gradient $\nabla L\left(w_1^{(k)}, w_2^{(k)}\right) = \left[\frac{\partial L}{\partial w_1^{(k)}}, \frac{\partial L}{\partial w_2^{(k)}}\right]^T$
- $\left(g_1^{(k)}, g_2^{(k)}\right) = \beta\left(g_1^{(k-1)}, g_2^{(k-1)}\right) + (1-\beta)\left(\left[\frac{\partial L}{\partial w_1^{(k)}}, \frac{\partial L}{\partial w_2^{(k)}}\right]^T\right)^2$ % Accumulate square gradients by EWMA
- $\left(w_1^{(k+1)}, w_2^{(k+1)}\right) = (w_1^{(k)}, w_2^{(k)}) - \alpha \frac{1}{\left(\sqrt{g_1^{(k)}+\varepsilon}, \sqrt{g_2^{(k)}+\varepsilon}\right)}\left[\frac{\partial L}{\partial w_1^{(k)}}, \frac{\partial L}{\partial w_2^{(k)}}\right]^T$ %Update position

%Small step size to frequently updated feature
%Large step size to not frequently updated feature

- $k = k + 1$;

EndWhile

End

Adam

Adam

%Inputs

$L(W)$: $\mathbb{R}^n \to \mathbb{R}$, a smooth loss function

$W^{(0)} \in \mathbb{R}^n$ The initial weight parameters

$M^{(0)} = 0_n \in \mathbb{R}^n$: Archive for Accumulation of EWMA of gradients % **Momentum** (First Moment)

$G^{(0)} = 0_n \in \mathbb{R}^n$: Archive for Accumulation of EWMA of square gradients % **RMSProp** (Second Moment)

$\alpha \in \mathbb{R}$: Learning rate

ε: a small positive value

$\beta_1 \in [0,1]$: Smoothing parameter for EWMA of gradients (most preferable for DL models, $\beta_1 = 0.9$)

$\beta_2 \in [0,1]$: Smoothing parameter for EWMA of square gradients (most preferable $\beta_2 = 0.99$)

%Output: $W^* \in \mathbb{R}^n$ optimal weight vector

Begin

$k \leftarrow 1$

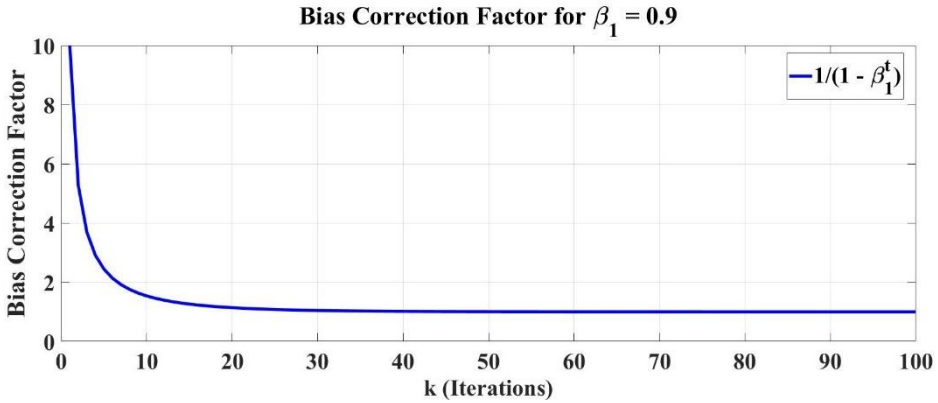
While (stopping criteria $\left\|\nabla_W L\left(W^{(k)}\right)\right\| > \varepsilon$) **do**

- Compute gradient $\nabla_W L\left(W^{(k)}\right)$
- $M^{(k)} = \beta_1 M^{(k-1)} + (1-\beta_1)\,\nabla_W L\left(W^{(k)}\right)$ % Accumulate gradients by **EWMA**
- $G^{(k)} = \beta_2 G^{(k-1)} + (1-\beta_2)\left(\nabla_W L\left(W^{(k)}\right)\right)^2$ % Accumulate square gradient by **EWMA**
- $\widehat{M}^{(k)} = \frac{1}{\left(1-\beta_1^k\right)} M^{(k)}$ % Bias corrected First Moment
- $\widehat{G}^{(k)} = \frac{1}{\left(1-\beta_2^k\right)} G^{(k)}$ % Bias corrected Second Moment
- $W^{(k)} = W^{(k-1)} - \frac{\alpha}{\sqrt{\widehat{G}^{(k)}+\varepsilon}} \widehat{M}^{(k)}$ %Update position
- $k = k + 1;$

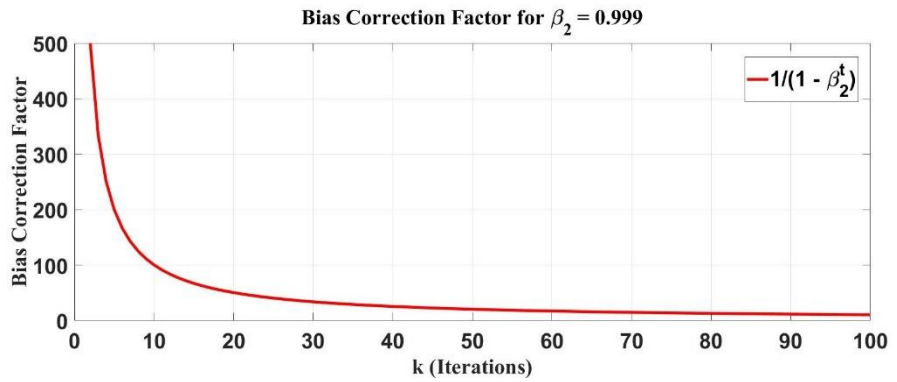
EndWhile

End

Bias correction:
Since $M^{(k)}$ (archive for the first moment) and $G^{(k)}$ (archive for the second moment) are initialized at zero. At the early stages of optimization (when k is small), these estimates are biased towards zero, especially when the smoothing parameters β_1 and β_2 are close to 1 (large value).



$\widehat{M}^{(k)} = \frac{1}{\left(1-\beta_1^k\right)} M^{(k)}$, as k increases, $\beta_1^k \to 0$ and $\frac{1}{\left(1-\beta_1^k\right)} \to 1$



$\widehat{G}^{(k)} = \frac{1}{\left(1-\beta_2^k\right)} G^{(k)}$, as k increases, $\beta_2^k \to 0$ and $\frac{1}{\left(1-\beta_2^k\right)} \to 1$

Bias correction:

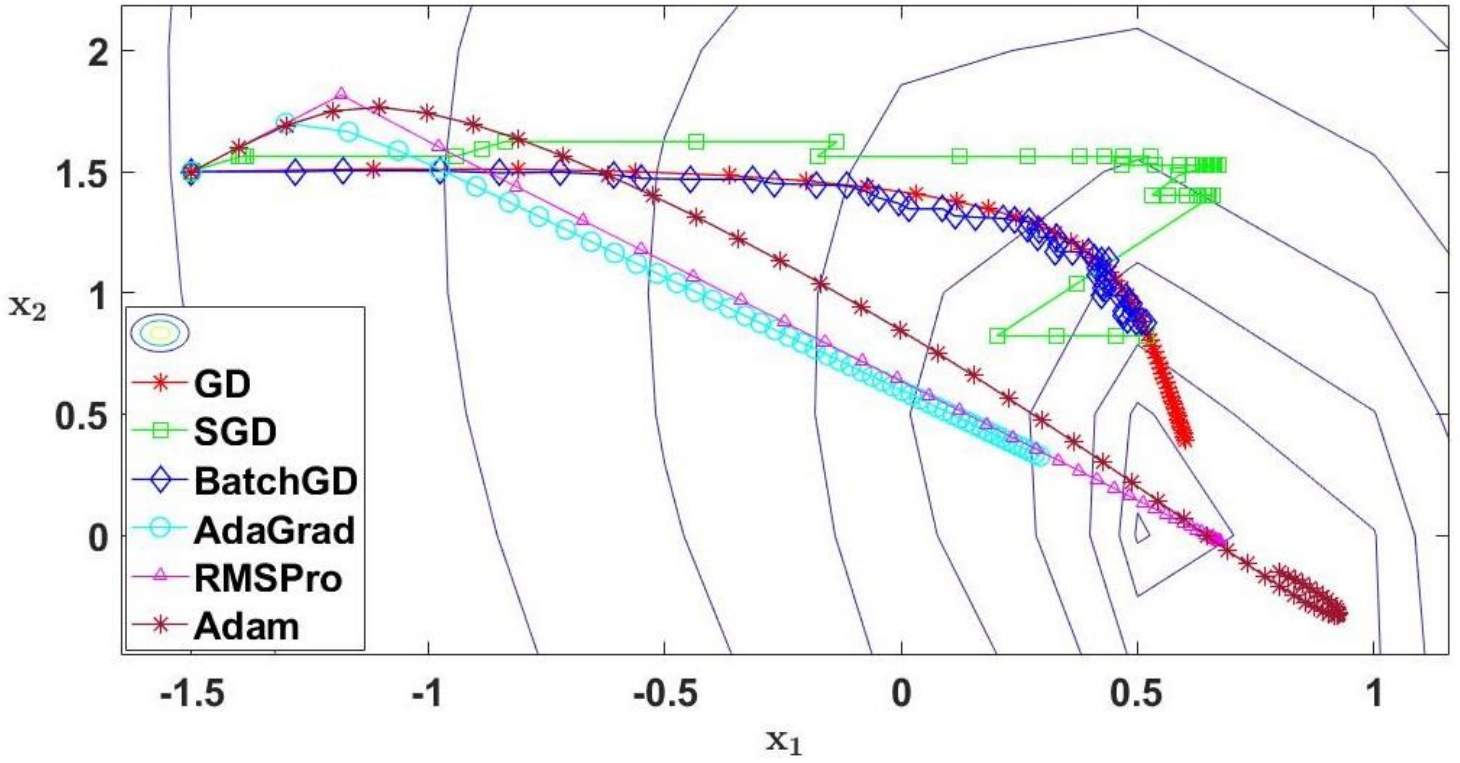
- **Bias correction:**
- **bias correction is needed because the first- and second-moment estimates are initialized at zero.**
  - $M^{(k)} = \beta_1 M^{(k-1)} + (1-\beta_1)\,\nabla_{\mathrm{W}} L\left(W^{(k)}\right)$ % Accumulate gradients by **EWMA**
  - $G^{(k)} = \beta_2 G^{(k-1)} + (1-\beta_2)\left(\nabla_{\mathrm{W}} L\left(W^{(k)}\right)\right)^2$ % Accumulate square gradient by **EWMA**
- **Initially,** $\boldsymbol{M^{(0)} = 0, G^{(0)} = 0,}$ $\beta_1 = 0.9, \beta_2 = 0.99$
- **At first iteration** $\boldsymbol{k = 1}$
  - $M^{(1)} = \beta_1 M^{(1-1)} + (1-\beta_1)\,\nabla_{\mathrm{W}} L\left(W^{(1)}\right)$
    - $= 0.9 \times 0 + (1-0.9)\nabla_{\mathrm{W}} L\left(W^{(1)}\right)$
    - $= 0.1\nabla_{\mathrm{W}} L\left(W^{(1)}\right)$
  - **So** $M^{(1)}$ **is much smaller than the actual gradient** $\nabla_{\mathrm{W}} L\left(W^{(1)}\right)$**).**
    - $G^{(k)} = \beta_2 G^{(k-1)} + (1-\beta_2)\left(\nabla_{\mathrm{W}} L\left(W^{(k)}\right)\right)^2$
    - $G^{(k)} = 0.99 \times 0 + (1-0.99)\left(\nabla_{\mathrm{W}} L\left(W^{(k)}\right)\right)^2$
      - $= 0.01 \times \left(\nabla_{\mathrm{W}} L\left(W^{(k)}\right)\right)^2$
- **Thus, because both moment estimates start from zero, they are systematically biased toward zero, particularly during the early iterations.**

Bias correction:

- **Bias correction:**
  - $\widehat{M}^{(k)} = \frac{1}{(1-\beta_1^k)} M^{(k)}$ % Bias corrected First Moment
  - $\widehat{G}^{(k)} = \frac{1}{(1-\beta_2^k)} G^{(k)}$ % Bias corrected Second Moment
- **At $t = 1$:**
  - $\widehat{M}^{(1)} = \frac{1}{(1-\beta_1^1)} M^{(1)} = \frac{1}{(1-0.9^1)} 0.1 \nabla_W L\left(W^{(1)}\right) = \nabla_W L\left(W^{(1)}\right)$
  - $\widehat{G}^{(1)} = \frac{1}{(1-\beta_2^1)} G^{(1)} = \frac{1}{(1-0.99^1)} 0.01 \times \left(\nabla_W L\left(W^{(k)}\right)\right)^2 = \left(\nabla_W L\left(W^{(k)}\right)\right)^2$

Adam uses bias correction because its exponential moving averages start from zero, causing the early estimates of the first and second moments to be biased toward zero.

Adam:



Limitations:

- Memory requirements
- Sensitivity to hyper-parameters
- Performance on non-convex problems
- Lack of theoretical understanding

Example

Optimize function $f(x) = x_1^2 + 2x_2^2$ by Adam optimizer up to 2 iterations with the following details:
Initial seed $x^{(0)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
Learning rate $\alpha = 0.01$
$\beta_1 = 0.9$
$\beta_1 = 0.99$
$\epsilon = 10^{-8}$

Apply the Adam optimizer to a linear regression problem with a two-dimensional feature space. Given the following data and parameters:
Feature matrix $X = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, Y = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$, bias term $b = 1$
$W^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
Learning rate $\alpha = 0.01$
$\beta_1 = 0.9$
$\beta_1 = 0.99$
$\epsilon = 10^{-8}$

Recall:

- **General linear regression model:**
- **General from:**
  - For $X \in \mathbb{R}^{N\times d}, W \in \mathbb{R}^{d}, Y \in \mathbb{R}^{N}, b \in \mathbb{R}$
  - Hypothesis (prediction): $\hat{Y} = XW + b$
  - Error: $E = \left(\hat{Y} - Y\right) = (XW + b - \mathrm{Y})$
  - MSE (Loss function): $L(W,b) = \frac{1}{2\times N}\sum_{i=1}^{N} E_i^2 = \frac{1}{2\times N}\|(XW+b) - Y\|^2$
  - $\nabla_W L(W,b) = \frac{1}{N}X^T(XW + b - Y) = \frac{1}{N}X^T E$
  - $\nabla_b L(W,b) = \frac{\partial L}{\partial b} = \frac{1}{N}1^T(XW + b - Y) = \frac{1}{N}1^T E$
- **Augmented form:**
  - $\tilde{X} = [1^{N\times 1}\ \ X] \in \mathbb{R}^{N\times(d+1)}$. Here $1^{N\times 1}$ is a column vector of ones (bias term).
  - $\widetilde{W} = \begin{bmatrix} b \\ W \end{bmatrix} \in \mathbb{R}^{(d+1)}$
  - Hypothesis (prediction): $\hat{Y} = \tilde{X}\widetilde{W}$
  - Error: $E = \left(\hat{Y} - Y\right) = (\tilde{X}\widetilde{W} - \mathrm{Y})$
  - MSE (Loss function): $L\left(\widetilde{W}\right) = \frac{1}{2\times N}\sum_{i=1}^{N} E_i^2 = \frac{1}{2\times N}\left\|\left(\tilde{X}\widetilde{W}\right) - Y\right\|^2$
  - $\nabla_{\widetilde{W}} L\left(\widetilde{W}\right) = \frac{1}{N}\tilde{X}^T(\tilde{X}\widetilde{W} - Y) = \frac{1}{N}\tilde{X}^T E$

- **General logistic regression model:**
- **General from:**
  - For $X \in \mathbb{R}^{N\times d}, W \in \mathbb{R}^{d}, Y \in \{0,1\}^{N}, b \in \mathbb{R}$
  - Hypothesis (prediction): $\hat{Y} = \sigma(XW + b)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$
  - Error: $E = \left(\hat{Y} - Y\right) = \sigma(XW + b) - \mathrm{Y}$
  - Binary cross entropy (Loss function):
  - $L(W,b) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)]$
  - $\nabla_W L(W,b) = \frac{1}{N}X^T(\sigma(XW + b) - Y) = \frac{1}{N}X^T E$
  - $\nabla_b L(W,b) = \frac{\partial L}{\partial b} = \frac{1}{N}1^T(\sigma(XW + b) - Y) = \frac{1}{N}1^T E$
- **Augmented form:**
  - $\tilde{X} = [1^{N\times 1}\ \ X] \in \mathbb{R}^{N\times(d+1)}$. Here $1^{N\times 1}$ is a column vector of ones (bias term).
  - $\widetilde{W} = \begin{bmatrix} b \\ W \end{bmatrix} \in \mathbb{R}^{(d+1)}$
  - Hypothesis (prediction): $\hat{Y} = \sigma(\tilde{X}\widetilde{W})$
  - Error: $E = \left(\hat{Y} - Y\right) = (\sigma(\tilde{X}\widetilde{W}) - \mathrm{Y})$
  - $L\left(\widetilde{W}\right) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)]$
  - $\nabla_{\widetilde{W}} L\left(\widetilde{W}\right) = \frac{1}{N}\tilde{X}^T(\sigma(\tilde{X}\widetilde{W}) - Y) = \frac{1}{N}\tilde{X}^T E$

Numerical example:

Apply the Adam optimizer to a linear regression problem with a two-dimensional feature space. Given the following data and parameters:

Feature matrix $X = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, Y = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$, bias term $b = 1$

$W^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Learning rate $\alpha = 0.01$

$\beta_1 = 0.9$

$\beta_1 = 0.99$

$\epsilon = 10^{-8}$

Step 1: Augment the Feature Matrix

- Since we have a bias term $b = 1$, we modify X to include a column of ones:
- Feature matrix $\tilde{X} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \end{bmatrix}$, Weight vector $\widetilde{W}^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, Target vector $Y = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$

- **Step 2: Define the Hypothesis Function**
- Compute prediction by linear regression model:
  - $\hat{Y}^{(0)} = \tilde{X}\widetilde{W}^{(0)} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 1 \times 0 + 2 \times 0 \\ 1 \times 1 + 2 \times 0 + 5 \times 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

- **General linear regression model:**
- **General from:**
  - For $X \in \mathbb{R}^{N \times d}, W \in \mathbb{R}^{d}, Y \in \mathbb{R}^{N}, b \in \mathbb{R}$
  - Hypothesis (prediction): $\hat{Y} = XW + b$
  - Error: $E = \left(\hat{Y} - Y\right) = (XW + b - \mathrm{Y})$
  - MSE (Loss function): $L(W, b) = \frac{1}{2 \times N} \sum_{i=1}^{N} E_i^2 = \frac{1}{2 \times N} \|(XW + b) - Y\|^2$
  - $\nabla_W L(W, b) = \frac{1}{N} X^T (XW + b - Y) = \frac{1}{N} X^T E$
  - $\nabla_b L(W, b) = \frac{\partial L}{\partial b} = \frac{1}{N} 1^T (XW + b - Y) = \frac{1}{N} 1^T E$
- **Augmented form:**
  - $\tilde{X} = [1^{N \times 1} \ X] \in \mathbb{R}^{N \times (d+1)}$. Here $1^{N \times 1}$ is a column vector of ones (bias term).
  - $\widetilde{W} = \begin{bmatrix} b \\ W \end{bmatrix} \in \mathbb{R}^{(d+1)}$
  - Hypothesis (prediction): $\hat{Y} = \tilde{X}\widetilde{W}$
  - Error: $E = \left(\hat{Y} - Y\right) = (\tilde{X}\widetilde{W} - \mathrm{Y})$
  - MSE (Loss function): $L\left(\widetilde{W}\right) = \frac{1}{2 \times N} \sum_{i=1}^{N} E_i^2 = \frac{1}{2 \times N} \left\|\left(\tilde{X}\widetilde{W}\right) - Y\right\|^2$
  - $\nabla_{\widetilde{W}} L\left(\widetilde{W}\right) = \frac{1}{N} \tilde{X}^T (\tilde{X}\widetilde{W} - Y) = \frac{1}{N} \tilde{X}^T E$

Numerical example: contd.

- **General linear regression model:**
- **Augmented form:**
  - $\tilde{X} = [1^{N\times 1}\ \ X] \in \mathbb{R}^{N\times(d+1)}$. Here $1^{N\times 1}$ is a column vector of ones (bias term).
  - $\widetilde{W} = \begin{bmatrix} b \\ W \end{bmatrix} \in \mathbb{R}^{(d+1)}$
  - Hypothesis (prediction): $\hat{Y} = \tilde{X}\widetilde{W}$
  - Error: $E = \left(\hat{Y} - Y\right) = (\tilde{X}\widetilde{W} - \mathrm{Y})$
  - MSE (Loss function): $L\left(\widetilde{W}\right) = \frac{1}{2\times N}\sum_{i=1}^{N} E_i^2 = \frac{1}{2\times N}\left\|\left(\tilde{X}\widetilde{W}\right) - Y\right\|^2$
  - $\nabla_{\widetilde{W}} L\left(\widetilde{W}\right) = \frac{1}{N}\tilde{X}^T(\tilde{X}\widetilde{W} - Y) = \frac{1}{N}\tilde{X}^T E$

- Error calculation:
  - $E^{(0)} = \left(\hat{Y}^{(0)} - Y\right) = \begin{bmatrix}1\\1\end{bmatrix} - \begin{bmatrix}3\\7\end{bmatrix} = \begin{bmatrix}-2\\-6\end{bmatrix}$

- Loss function value: MSE value
- $L\left(\widetilde{W}^{(0)}\right) = \frac{1}{2}\left\|E^{(0)}\right\|^2 = \left(\sqrt{(-2)^2 + (-6)^2}\right)^2 = \frac{1}{2}(4 + 36) = 20$

- Compute gradient
  - $g^{(0)} = \nabla_{\widetilde{W}^{(0)}} L\left(\widetilde{W}^{(0)}\right) = \tilde{X}^T\left(\tilde{X}\widetilde{W}^{(0)} - Y\right) = \tilde{X}^T E^{(0)} = \begin{bmatrix}1 & 1\\ 1 & 2\\ 2 & 5\end{bmatrix}\begin{bmatrix}-2\\-6\end{bmatrix} = \begin{bmatrix}1\times -2 + 1\times -6\\ 1\times -2 + 2\times -6\\ 2\times -2 + 5\times -6\end{bmatrix} = \begin{bmatrix}-8\\-14\\-34\end{bmatrix}$

Initialize Adam Variables:

- $M^{(0)} = (0,0,0)^T$: Archive for Accumulation of EWMA of gradients % Momentum (First Moment)
- $G^{(0)} = (0,0,0)^T$: : Archive for Accumulation of EWMA of square gradients % RMSProp (Second Moment)
- $\alpha = 0.01, \beta_1 = 0.9, \beta_1 = 0.99$

Numerical example: contd.

Adam update $k = 1$:

- **Update First Moment estimate:**
  - $M^{(1)} = \beta_1 M^{(0)} + (1-\beta_1)\, g^{(0)}$
  - $M^{(1)} = 0.9 \times \begin{bmatrix}0\\0\\0\end{bmatrix} + (1-0.9)\begin{bmatrix}-8\\-14\\-34\end{bmatrix} = (0.1)\begin{bmatrix}-8\\-14\\-34\end{bmatrix} = \begin{bmatrix}-0.8\\-1.4\\-3.4\end{bmatrix}$
- **Update second Moment estimate:**
  - $G^{(1)} = \beta_2 G^{(0)} + (1-\beta_2)\left(g^{(0)}\right)^2$
  - $G^{(1)} = 0.99 \times \begin{bmatrix}0\\0\\0\end{bmatrix} + (1-0.99)\left(\begin{bmatrix}-8\\-14\\-34\end{bmatrix}\right)^2 = (0.01)\begin{bmatrix}64\\196\\1156\end{bmatrix} = \begin{bmatrix}0.64\\1.96\\11.56\end{bmatrix}$
- **Bias Correction:**
  - $\widehat{M}^{(1)} = \frac{1}{\left(1-\beta_1^1\right)} M^{(1)} = \frac{1}{(1-0.9^1)}\begin{bmatrix}-0.8\\-1.4\\-3.4\end{bmatrix} = \frac{1}{(0.1)}\begin{bmatrix}-0.8\\-1.4\\-3.4\end{bmatrix} = \begin{bmatrix}-8\\-14\\-34\end{bmatrix}$
  - $\widehat{G}^{(1)} = \frac{1}{\left(1-\beta_2^1\right)} G^{(1)} = \frac{1}{(1-0.99^1)}\begin{bmatrix}0.64\\1.96\\11.56\end{bmatrix} = \frac{1}{(0.01)}\begin{bmatrix}0.64\\1.96\\11.56\end{bmatrix} = \begin{bmatrix}64\\196\\1156\end{bmatrix}$

- **Compute Update Step:**
  - $\widetilde{W}^{(1)} = \widetilde{W}^{(0)} - \frac{\alpha}{\sqrt{\widehat{G}^{(1)}+\varepsilon}}\widehat{M}^{(1)}$
  - **Calculate** $\sqrt{\widehat{G}^{(1)}} + \epsilon = \begin{bmatrix}\sqrt{64}+\varepsilon\\\sqrt{196}+\varepsilon\\\sqrt{1156}+\varepsilon\end{bmatrix} \approx \begin{bmatrix}8\\14\\34\end{bmatrix}$
  - **Calculate** $\frac{1}{\sqrt{\widehat{G}^{(1)}}+\epsilon} \approx \begin{bmatrix}\frac{1}{8}\\\frac{1}{14}\\\frac{1}{34}\end{bmatrix}$
  - **Calculate** $\frac{\alpha}{\sqrt{\widehat{G}^{(1)}+\varepsilon}} = \begin{bmatrix}0.01\times\frac{1}{8}\\0.01\times\frac{1}{14}\\0.01\times\frac{1}{34}\end{bmatrix} = \begin{bmatrix}0.00125\\0.000714\\0.000294\end{bmatrix}$
  - **Calculate** $\frac{\alpha}{\sqrt{\widehat{G}^{(1)}+\varepsilon}}\widehat{M}^{(1)} = \begin{bmatrix}0.00125\\0.000714\\0.000294\end{bmatrix} \odot \begin{bmatrix}-8\\-14\\-34\end{bmatrix} = \begin{bmatrix}-0.01\\-0.009996\\-0.009996\end{bmatrix}$
  - Finally,
  - $\widetilde{W}^{(1)} = \widetilde{W}^{(0)} - \frac{\alpha}{\sqrt{\widehat{G}^{(1)}+\varepsilon}}\widehat{M}^{(1)} \Rightarrow \widetilde{W}^{(1)} = \begin{bmatrix}1\\0\\0\end{bmatrix} - \begin{bmatrix}-0.01\\-0.009996\\-0.009996\end{bmatrix} = \begin{bmatrix}1.01\\0.009996\\0.009996\end{bmatrix}$

- **Compute prediction with $\widetilde{W}^{(1)}$:**
- $\hat{Y}^{(1)} = \tilde{X}\widetilde{W}^{(1)} = \begin{bmatrix}1&1&2\\1&2&5\end{bmatrix}\begin{bmatrix}1.01\\0.009996\\0.009996\end{bmatrix} = \begin{bmatrix}1(1.01)+1(0.009996)+1(0.009996)\\1(1.01)+2(0.009996)+5(0.009996)\end{bmatrix} = \begin{bmatrix}1.029992\\1.079972\end{bmatrix}$

- Error calculation:
  - $E^{(1)} = \left(\hat{Y}^{(1)} - Y\right) = \begin{bmatrix}1.029992\\1.079972\end{bmatrix} - \begin{bmatrix}3\\7\end{bmatrix} = \begin{bmatrix}-1.97008\\-5.920028\end{bmatrix}$

- Compute loss:
  - $L\left(\widetilde{W}^{(1)}\right) = \frac{1}{2}\left\|E^{(1)}\right\|^2 = \frac{1}{2}[(-1.97008)^2+(-5.920028)^2] = 19.46$

Numerical example: contd.

Adam update $k = 2$

- Compute gradient
  - $g^{(1)} = \nabla_{\widetilde{W}^{(1)}} L\left(\widetilde{W}^{(1)}\right) = \tilde{X}^T\left(\tilde{X}\widetilde{W}^{(1)} - Y\right) = \tilde{X}^T E^{(1)} = \begin{bmatrix}1 & 1\\ 1 & 2\\ 2 & 5\end{bmatrix}\begin{bmatrix}-1.97008\\ -5.920028\end{bmatrix} = \begin{bmatrix}1(-1.970008)+1(-5.920028)\\ 1(-1.970008)+2(-5.920028)\\ 1(-1.970008)+5(-5.920028)\end{bmatrix} = \begin{bmatrix}-1.970008-5.920028\\ -1.970008-11.840056\\ -1.970008-29.600140\end{bmatrix} = \begin{bmatrix}-7.890036\\ -13.810064\\ -31.570148\end{bmatrix}$

- **Update First Moment estimate:**
  - $M^{(2)} = \beta_1 M^{(1)} + (1-\beta_1)\, g^{(1)}$
  - $M^{(2)} = 0.9 \times \begin{bmatrix}-0.8\\ -1.4\\ -3.4\end{bmatrix} + (1-0.9)\begin{bmatrix}-7.890036\\ -13.810064\\ -31.570148\end{bmatrix} = \begin{bmatrix}-1.509\\ -2.641\\ -6.197\end{bmatrix}$
- **Update second Moment estimate:**
  - $G^{(2)} = \beta_2 G^{(1)} + (1-\beta_2)\left(g^{(1)}\right)^2$
  - $G^{(2)} = 0.99 \times \begin{bmatrix}0.64\\ 1.96\\ 11.56\end{bmatrix} + (1-0.99)\left(\begin{bmatrix}-7.890036\\ -13.810064\\ -31.570148\end{bmatrix}\right)^2 = 0.99 \times \begin{bmatrix}0.64\\ 1.96\\ 11.56\end{bmatrix} + (0.01)\begin{bmatrix}62.28\\ 190.71\\ 996.82\end{bmatrix} = \begin{bmatrix}1.2528\\ 3.8291\\ 21.4926\end{bmatrix}$
- **Bias Correction:**
  - $\widehat{M}^{(2)} = \frac{1}{\left(1-\beta_1^2\right)} M^{(2)} = \frac{1}{(1-0.9^2)}\begin{bmatrix}-1.509\\ -2.641\\ -6.197\end{bmatrix} = \frac{1}{(0.19)}\begin{bmatrix}-1.509\\ -2.641\\ -6.197\end{bmatrix} = \begin{bmatrix}-7.94\\ -13.90\\ -32.61\end{bmatrix}$
  - $\widehat{G}^{(2)} = \frac{1}{\left(1-\beta_2^2\right)} G^{(2)} = \frac{1}{(1-0.99^2)}\begin{bmatrix}1.2528\\ 3.8291\\ 21.4926\end{bmatrix} = \frac{1}{(0.0199)}\begin{bmatrix}1.2528\\ 3.8291\\ 21.4926\end{bmatrix} = \begin{bmatrix}62.96\\ 192.51\\ 1080.93\end{bmatrix}$

Numerical example: contd.

- **Compute Update Step:**
  - $\widetilde{W}^{(2)} = \widetilde{W}^{(1)} - \frac{\alpha}{\sqrt{\hat{G}^{(2)}+\varepsilon}}\widehat{M}^{(2)}$
  - **Calculate** $\sqrt{\hat{G}^{(2)}} + \epsilon = \begin{bmatrix} \sqrt{62.96}+\varepsilon \\ \sqrt{192.51}+\varepsilon \\ \sqrt{1080.93}+\varepsilon \end{bmatrix} \approx \begin{bmatrix} 7.94 \\ 13.88 \\ 32.89 \end{bmatrix}$
  - **Calculate** $\frac{\mathbf{1}}{\sqrt{\hat{G}^{(2)}}+\epsilon} \approx \begin{bmatrix} \frac{1}{7.94} \\ \frac{1}{13.88} \\ \frac{1}{32.89} \end{bmatrix} \approx \begin{bmatrix} 0.126 \\ 0.072 \\ 0.030 \end{bmatrix}$
  - **Calculate** $\frac{\alpha}{\sqrt{\hat{G}^{(2)}+\varepsilon}} = \begin{bmatrix} 0.01 \times 0.126 \\ 0.01 \times 0.072 \\ 0.01 \times 0.030 \end{bmatrix} = \begin{bmatrix} 0.00126 \\ 0.00072 \\ 0.00030 \end{bmatrix}$
  - **Calculate** $\frac{\alpha}{\sqrt{\hat{G}^{(2)}+\varepsilon}}\widehat{M}^{(2)} = \begin{bmatrix} 0.00126 \\ 0.00072 \\ 0.00030 \end{bmatrix} \odot \begin{bmatrix} -7.94 \\ -13.90 \\ -32.61 \end{bmatrix} = \begin{bmatrix} 0.00126\times(-7.94) \\ 0.00072\times(-13.90) \\ 0.00030\times(-32.61) \end{bmatrix} = \begin{bmatrix} -0.01000 \\ -0.01000 \\ -0.00978 \end{bmatrix}$
  - Finally,
  - $\widetilde{W}^{(2)} = \widetilde{W}^{(1)} - \frac{\alpha}{\sqrt{\hat{G}^{(2)}+\varepsilon}}\widehat{M}^{(2)} \Rightarrow \widetilde{W}^{(1)} = \begin{bmatrix} 1.01 \\ 0.009996 \\ 0.009996 \end{bmatrix} - \begin{bmatrix} -0.01000 \\ -0.01000 \\ -0.00978 \end{bmatrix} = \begin{bmatrix} 1.02 \\ 0.019996 \\ 0.019776 \end{bmatrix}$

- **Compute prediction with $\widetilde{W}^{(2)}$:**
- $\hat{Y}^{(2)} = \tilde{X}\widetilde{W}^{(2)} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1.02 \\ 0.019996 \\ 0.019776 \end{bmatrix} = \begin{bmatrix} 1(1.02)+1(0.019996)+2(0.019776) \\ 1(1.02)+2(0.019996)+5(0.019776) \end{bmatrix} = \begin{bmatrix} 1.079544 \\ 1.158872 \end{bmatrix}$

- Error calculation:
  - $E^{(2)} = \left(\hat{Y}^{(2)} - Y\right) = \begin{bmatrix} 1.079544 \\ 1.158872 \end{bmatrix} - \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} -1.920456 \\ -5.841128 \end{bmatrix}$

- Compute loss:
  - $L\left(\widetilde{W}^{(2)}\right) = \frac{1}{2}\left\|E^{(2)}\right\|^2 = \frac{1}{2}[(-1.920456)^2+(-5.841128)^2] = 18.9174$

Optimize function $f(x) = x_1^2 + 2x_2^2$ by Adam optimizer up to 2 iterations with the following details:

Initial seed $x^{(0)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Learning rate $\alpha = 0.01$

$\beta_1 = 0.9$

$\beta_1 = 0.99$

$\epsilon = 10^{-8}$

- Step1 : Compute the gradient:

$$\nabla_x f(x) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 4x_2 \end{bmatrix}$$

- At $x^{(0)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$,
- $\nabla_{x^{(0)}} f(x) = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$
- Objective value $f\left(x^{(0)}\right) = 1^2 + 2(-1)^2 = 1 + 2 = 3$

Initialize Adam Variables:

- $M^{(0)} = (0,0\ )^T$: Archive for Accumulation of EWMA of gradients % Momentum (First Moment)
- $G^{(0)} = (0,0)^T$: : Archive for Accumulation of EWMA of square gradients % RMSProp (Second Moment)

$\alpha = 0.01, \beta_1 = 0.9, \beta_1 = 0.99$

- 1

Adam update $k = 1$:

- **Update First Moment estimate:**
  - $M^{(1)} = \beta_1 M^{(0)} + (1 - \beta_1)\, g^{(0)}$
  - $M^{(1)} = 0.9 \times \begin{bmatrix}0\\0\end{bmatrix} + (1 - 0.9) \begin{bmatrix}2\\-4\end{bmatrix} = \begin{bmatrix}0.2\\-0.4\end{bmatrix}$
- **Update second Moment estimate:**
  - $G^{(1)} = \beta_2 G^{(0)} + (1 - \beta_2)\left(g^{(0)}\right)^2$
  - $G^{(1)} = 0.99 \times \begin{bmatrix}0\\0\end{bmatrix} + (1 - 0.99) \left(\begin{bmatrix}2\\-4\end{bmatrix}\right)^2 = \begin{bmatrix}0.04\\0.16\end{bmatrix}$
- **Bias Correction:**
  - $\widehat{M}^{(1)} = \frac{1}{\left(1-\beta_1^1\right)} M^{(1)} = \frac{1}{(1-0.9^1)} \begin{bmatrix}0.2\\-0.4\end{bmatrix} = \frac{1}{(0.1)} \begin{bmatrix}0.2\\-0.4\end{bmatrix} = \begin{bmatrix}2\\-4\end{bmatrix}$
  - $\widehat{G}^{(1)} = \frac{1}{\left(1-\beta_2^1\right)} G^{(1)} = \frac{1}{(1-0.99^1)} \begin{bmatrix}0.04\\0.16\end{bmatrix} = \frac{1}{(0.01)} \begin{bmatrix}0.04\\0.16\end{bmatrix} = \begin{bmatrix}4\\16\end{bmatrix}$

- **Compute Update Step:**
  - $x^{(1)} = x^{(0)} - \frac{\alpha}{\sqrt{\widehat{G}^{(1)}+\varepsilon}} \widehat{M}^{(1)}$
  - $x^{(1)} = \begin{bmatrix}1\\-1\end{bmatrix} - 0.01 \times \frac{\begin{bmatrix}2\\-4\end{bmatrix}}{\begin{bmatrix}\sqrt{4}+10^{-8}\\\sqrt{16}+10^{-8}\end{bmatrix}}$
  - $x^{(1)} = \begin{bmatrix}1\\-1\end{bmatrix} - 0.01 \times \begin{bmatrix}1\\-1\end{bmatrix}$
  - $x^{(1)} = \begin{bmatrix}0.99\\-0.99\end{bmatrix}$

- Objective value $f\left(x^{(1)}\right) = 0.99^2 + 2(-0.99)^2 = 2.94$

Adam update $k = 2$
Continue the same process: